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% Problem 5: Feedforward-Feedback Loop Simulation

```
clear; clc;
```

```
s = tf('s');
```

% Parameters

```
Kp = 2;
```

```
tau_p = 95;
```

```
theta_p = 20;
```

```
Kd = 0.5;
```

```
tau_d = 45;
```

```
theta_d = 20;
```

% Pade approximations for delays (1st order)

```
Gp = Kp * pade(exp(-theta_p*s), 1) / (tau_p*s + 1);
```

```
Gd = Kd * pade(exp(-theta_d*s), 1) / (tau_d*s + 1);
```

% (a) Feedforward controller (ideal cancellation, Gd/Gp)

```
Gff = -Kd * (tau_p*s + 1) / (Kp * (tau_d*s + 1));
```

% (b) Feedback controller design with tau_c = 30

```
tau_c = 30;
```

```
Kc = (tau_p / (Kp * (tau_c + theta_p)));
```

```
tauI = tau_p;
```

```
Gc = Kc * (1 + 1/(tauI*s)); % PI Controller
```

% (c) Feedback-only response to disturbance step

```
T_fb = feedback(Gd, Gc*Gp);
```

```
figure(1);
```

```
step(T_fb, 200);
```

```
title('Figure 1: Feedback-only Step Response to Disturbance');
```

```
xlabel('Time (s)');
```

```
ylabel('y(t)');
```

```
grid on;
```

% (d) Feedforward + Feedback response (correct Gd used)

```
numerator_ff = minreal(Gd + Gff*Gp);
```

```
T_ff_fb = feedback(numerator_ff, Gc*Gp);
```

```
figure(2);
```

```
step(T_ff_fb, 200);
```

```
title('Figure 2: Feedforward + Feedback Response (Correct Gd)');
```

```
xlabel('Time (s)');
```

```
ylabel('y(t)');
```

```

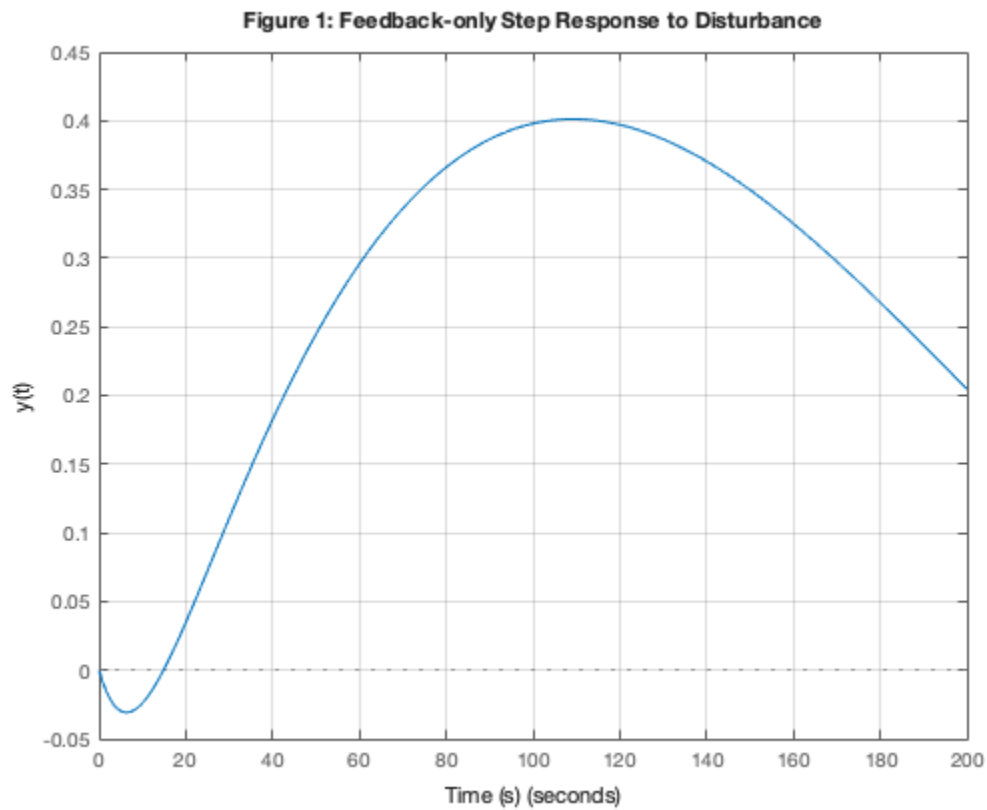
grid on;

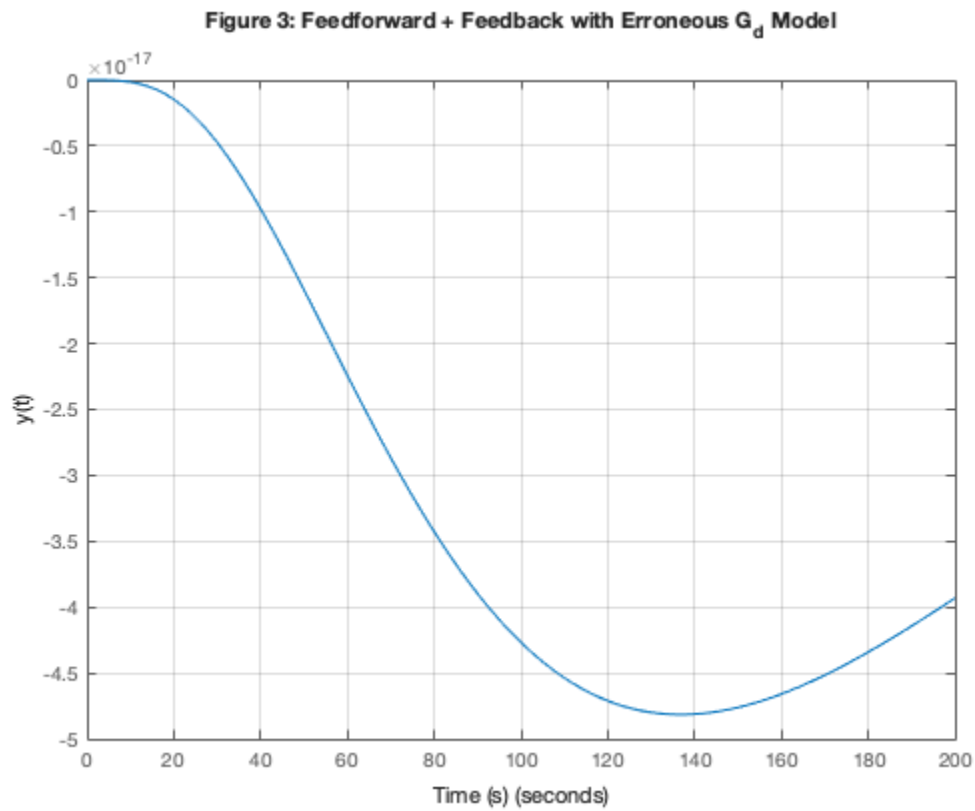
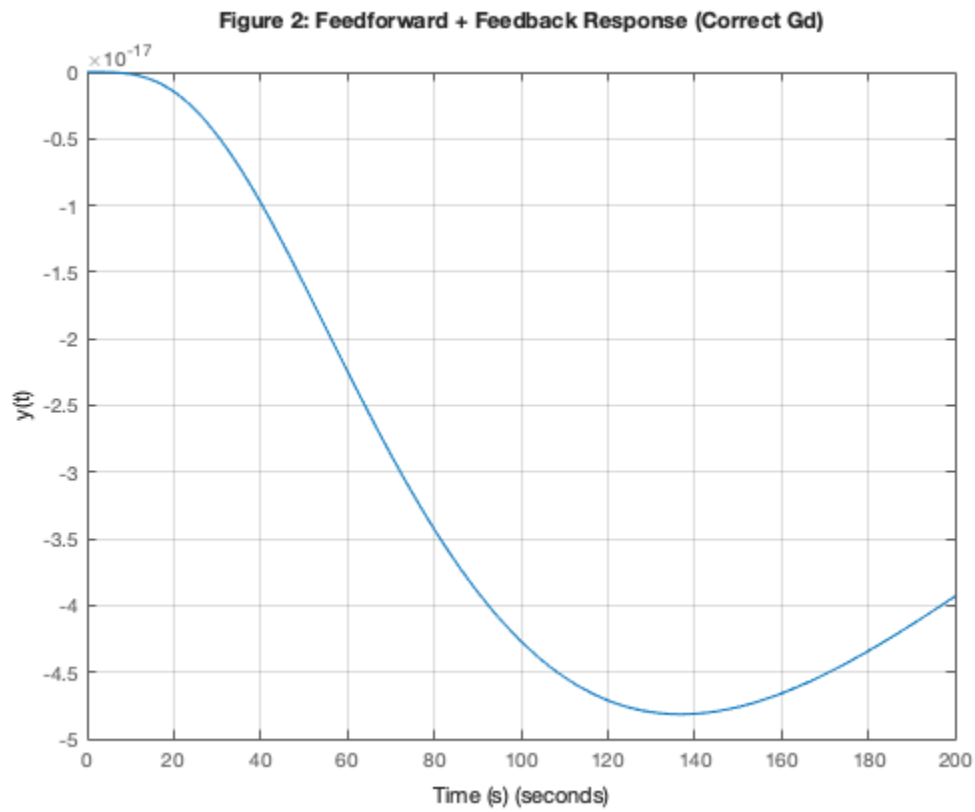
% (e) Erroneous disturbance model used in feedforward design
Gd_wrong = 0.5 * pade(exp(-30*s), 1) / (60*s + 1);
Gff_wrong = -0.25 * (tau_p*s + 1) / (tau_d*s + 1); % Based on incorrect Gd

numerator_wrong = minreal(Gd + Gff_wrong*Gp); % Real (true Gd)
T_ff_fb_wrong = feedback(numerator_wrong, Gc*Gp);

figure(3);
step(T_ff_fb_wrong, 200);
title('Figure 3: Feedforward + Feedback with Erroneous G_d Model');
xlabel('Time (s)');
ylabel('y(t)');
grid on;

```





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